

Spencer H. Bryngelson

I Basic information

Title: Assistant Professor, School of Computational Science & Engineering
Assistant Professor by Courtesy (o%), Daniel Guggenheim School of Aerospace Engineering
Assistant Professor by Courtesy (o%), George W. Woodruff School of Mechanical Engineering
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2 Education

- University of Illinois at Urbana–Champaign
(2018) Doctor of Philosophy, Theoretical & Applied Mechanics
Advisor: Jonathan Freund
(2015) Master of Science, Theoretical & Applied Mechanics
- University of Michigan–Dearborn
(2013) Bachelor of Science, Mechanical Engineering
(2013) Bachelor of Science, Engineering Mathematics

3 Positions held

- (2021–Present) Assistant Professor, School of Computational Science & Engineering, Georgia Institute of Technology
- (2023–Present) Assistant Professor by Courtesy (o%), Daniel Guggenheim School of Aerospace Engineering, College of Engineering, Georgia Institute of Technology
- (2024–Present) Assistant Professor by Courtesy (o%), George W. Woodruff School of Mechanical Engineering, College of Engineering, Georgia Institute of Technology
- (2022) Visiting Scholar, Stanford University, Center for Turbulence Research (Summer Program)
- (2018–21) Senior Postdoctoral Scholar, California Institute of Technology, with Tim Colonius
- (2019) Visiting Researcher, Massachusetts Institute of Technology, with Themis Sapsis
- (2018) Postdoctoral Researcher, XPACC (PSAAP II center), with Carlos Pantano, Dan Bodony, Jon Freund

4 Research support

4.1 Pending grants

- (2024–29) PI: NSF CAREER “*From Remodeling to Stroke, Vasculature to Cell: Unraveling the mechanics of sickle cell disease*” (\$544K)
- (2025–28) co-PI: NSF CSSI “*Elements: High-performance algorithm modules with complexity improvements for PDEs in computational physics*” (\$480K, SHB share: \$240K)
- (2024–27) PI: DOD AFOSR YIP “*Actuator design and control in discontinuous flow fields*” (\$450K)

4.2 Funded grants

- (2024–29) Senior personnel: DARPA Young Faculty Award “*Squid-inspired nozzles for enhanced efficiency and thrust in rotary propulsors*” (\$1M), PI: S. Bhamla (GT), SHB Share: \$150K Y1–2
- (2024–27) co-PI: DOD ONR N00014-24-1-2094 “*Multi-scale simulations of combustion in a solid propellant ramjet with embedded reactive metal particles*” (\$375K), PI: S. Menon (GT), SHB Share: \$125K
- (2024) PI: DOE DE-AC52-07NA27344 (Lawrence Livermore National Laboratories subcontract), “*Accelerated, Compressed, and Regularized Computation of Kinetic-based PDEs*” (\$80K)
- (2023–28) co-PI: DOD ONR MURI N00014-23-1-2501, “*Combustion of solid fuels in high enthalpy flow*” (\$3.8M) PI: G. Young (Virginia Polytechnic Institute and State University), 7 other co-PIs. SHB Share: \$270K Y1–3
- (2023–27) PI: DOD ARO W911NF-23-10324, “*Investigation and inference of soft material deformation mechanisms unlocked at large speeds, finite deformations, and many cycles*,” collaborative with University of Michigan, Jon Estrada. (Total: \$835K; SHB Share: \$314K)

- (2023–24) PI: DOE DE-NA0003525 (Sandia National Laboratories subcontract), “*Vibrated bubbly flow simulation*” (\$100K)
- (2023–24) co-PI: DARPA HR0011-23-3-0006, “*Quantum eigensolvers in fluid-dynamic computations and applications*” (\$300K), PI B. Gard (Georgia Tech Research Institute), SHB Share: \$100K
- (2022–23) PI: DOE DE-NA0003525 (Sandia National Laboratories subcontract), “*Vibrated bubbly flow simulation*” (\$65K)
- (2022–23) PI: DOE ORAU Powe, “*A methodologically coherent multi-scale model for multiphase flow*” (\$10K)
- (2022–26) PI: DOD ONR N00014-22-12519, “*Stochastic framework for cavitating flows: mesoscale modeling and acceleration*” (\$560K)
- (2022–23) co-PI: GTRI IRAD, “*Quantum optimization for lattice Boltzmann simulation (QOLBS)*” (\$40K), PI: B. Gard (Georgia Tech Research Institute)
- (2022) PI: GT Seed Grant, Forming Teams “*Quantum computing for next-generation engineering simulation*” (\$50K)
- (2022) PI: GTQA DE-00013211, “*Quantum algorithms for lattice Boltzmann fluid flow simulation*” (\$14.5K)

5 Publications

5.1 Preprints

- [U4] Bryngelson, S. H. (2026). “ANEForge: Python for direct computation on the Apple Neural Engine”. arXiv:2606.17090. DOI: [10.48550/arXiv.2606.17090](https://doi.org/10.48550/arXiv.2606.17090).
- [U3] Chu, T., J. Beckett, Z. Zhu, J. B. Estrada, and S. H. Bryngelson (2026). “Limits of constant-parameter constitutive models for hydrogels under inertial cavitation”. arXiv:2606.13584. DOI: [10.48550/arXiv.2606.13584](https://doi.org/10.48550/arXiv.2606.13584).
- [U2] Chu, T., B. Wilfong, T. Koehler, R. M. McMullen, and S. H. Bryngelson (2026). “Coupled Rayleigh–Taylor and Faraday instabilities in vertically vibrated cylindrical containers”. arXiv:2607.28932. DOI: [10.48550/arXiv.2607.28932](https://doi.org/10.48550/arXiv.2607.28932).
- [U1] Radhakrishnan, A. and S. H. Bryngelson (2026). “A sub-grid-scale model for polydisperse bubbly flows with heat and mass transfer”. arXiv:2607.27426. DOI: [10.48550/arXiv.2607.27426](https://doi.org/10.48550/arXiv.2607.27426).

5.2 Archival, heavily refereed papers

- [P47] Bryngelson, S. H. (2026). “Fast integration method for averaging polydisperse bubble population dynamics”. *Computers & Fluids* **304**, 106877. DOI: [10.1016/j.compfluid.2025.106877](https://doi.org/10.1016/j.compfluid.2025.106877).
- [P46] Bryngelson, S. H., R. O. Fox, and F. Laurent (2026). “Fourth-order HyQMOM closures for multidimensional kinetic equations”. *Journal of Computational Physics* **566**, 115242. DOI: [10.1016/j.jcp.2026.115242](https://doi.org/10.1016/j.jcp.2026.115242).
- [P45] Chu, T., J. B. Estrada, and S. H. Bryngelson (2026). “Accelerating Bayesian optimal experimental design via local radial basis functions: Application to soft material characterization”. *Journal of Computational Physics* **562**, 115002. DOI: [10.1016/j.jcp.2026.115002](https://doi.org/10.1016/j.jcp.2026.115002).
- [P44] Cisneros-Garibay, Esteban, H. Le Berre, D. Adam, Spencer H. Bryngelson, and Jonathan B. Freund (2026). “Pyrometheus: Symbolic abstractions for XPU and automatically differentiated computation of combustion kinetics and thermodynamics”. *Computer Physics Communications* **320**, 109987. DOI: [10.1016/j.cpc.2025.109987](https://doi.org/10.1016/j.cpc.2025.109987).
- [P43] Goldstein, Bailey I., David N. Ramsey, Haocheng Yu, Spencer H. Bryngelson, and K. K. Ahuja (2026). “Performance of Sharply Bent Acoustic Resonators at High Sound Levels”. *ALAA Journal* **64** 8, 4719–4728. DOI: [10.2514/1.J066497](https://doi.org/10.2514/1.J066497).
- [P42] Jawetz, C. L., Z. Song, A. Alexeev, and S. H. Bryngelson (2026). “Quantum lattice Boltzmann algorithm for heat transfer with phase change”. *Quantum Science and Technology* **11** 3, 035032. DOI: [10.1088/2058-9565/ae7b7c](https://doi.org/10.1088/2058-9565/ae7b7c).
- [P41] Lee, M., Z. Song, S. Kocherla, A. Adams, A. Alexeev, and S. H. Bryngelson (2026). “A multiple-circuit approach to quantum resource reduction with application to the quantum lattice Boltzmann method”. *Future Generation Computing Systems* **174**, 107975. DOI: [10.1016/j.future.2025.107975](https://doi.org/10.1016/j.future.2025.107975).
- [P40] Radhakrishnan, A., B. Wilfong, S. H. Bryngelson, and F. Schäfer (2026). “Shocks without shock capturing: Information geometric regularization of finite volume methods for Navier-Stokes-like problems”. *Computer Methods in Applied Mechanics and Engineering* **461**, 119216. DOI: [10.1016/j.cma.2026.119216](https://doi.org/10.1016/j.cma.2026.119216).

- [P39] Sanchez, Victor, Sawyer Remillard, Bachir A. Abeid, Lehu Bu, Spencer H. Bryngelson, Jin Yang, Jonathan B. Estrada, and Mauro Rodriguez Jr. (2026). “Hierarchical Bayesian constitutive model selection for high-strain-rate soft material characterization”. *Soft Matter* **22**, 3207–3223. DOI: [10.1039/D5SM01193K](https://doi.org/10.1039/D5SM01193K).
- [P38] Song, Z., H. Ren, M. Lee, B. Gard, N. Renaud, and S. H. Bryngelson (2026). “Hadamard random forest for reconstructing real valued pure quantum states”. *Communications Physics*. DOI: [10.1038/s42005-026-02717-1](https://doi.org/10.1038/s42005-026-02717-1).
- [P37] Vaca-Revelo, D., B. Wilfong, S. H. Bryngelson, and A. Gnanaskandan (2026). “Hardware-Accelerated Phase-Averaging for Cavitating Bubbly Flows”. *International Journal of Multiphase Flow* **199**, 105674. DOI: [10.1016/j.ijmultiphaseflow.2026.105674](https://doi.org/10.1016/j.ijmultiphaseflow.2026.105674).
- [P36] Wilfong, Benjamin, Henry Le Berre, Anand Radhakrishnan, Ansh Gupta, Daniel J. Vickers, Diego Vaca-Revelo, Dimitrios Adam, Haocheng Yu, Hyeoksu Lee, Jose Rodolfo Chreim, Mirelys Carcana Barbosa, Yanjun Zhang, Esteban Cisneros-Garibay, Aswin Gnanaskandan, Mauro Rodriguez Jr., Reuben D. Budiardja, Stephen Abbott, Tim Colonius, and Spencer H. Bryngelson (2026). “MFC 5.0: An exascale many-physics flow solver”. *Computer Physics Communications* **322**, 110055. DOI: [10.1016/j.cpc.2026.110055](https://doi.org/10.1016/j.cpc.2026.110055).
- [P35] Yu, H., K. Ahuja, L. L. Sankar, and S. H. Bryngelson (2026). “Transmission of High-Amplitude Sound through Leakages of Ill-fitting Earplugs”. *International Journal of Aeroacoustics*. In press. DOI: [10.1177/1475472X261473636](https://doi.org/10.1177/1475472X261473636).
- [P34] *Yu, H., T. *Chu, and S. H. Bryngelson (2026). “Energy dissipation mechanisms in an acoustically-driven slit”. *Journal of Fluid Mechanics*. In press, *Equal contribution. DOI: [10.48550/arXiv.2512.19507](https://doi.org/10.48550/arXiv.2512.19507).
- [P33] Chu, T., J. B. Estrada, and S. H. Bryngelson (2025). “Bayesian optimal design accelerates discovery of soft material properties from bubble dynamics”. *Computational Mechanics* **76**, 431–447. DOI: [10.1007/s00466-025-02606-4](https://doi.org/10.1007/s00466-025-02606-4).
- [P32] Chu, T., B. Wilfong, T. Koehler, R. M. McMullen, and S. H. Bryngelson (2025). “Competing mechanisms at vibrated interfaces of density-stratified fluids”. *Physical Review Fluids* **10**, 093904. DOI: [10.1103/r9b3-psg4](https://doi.org/10.1103/r9b3-psg4).
- [P31] Song, Z., R. Deaton, B. Gard, and S. H. Bryngelson (2025). “Incompressible Navier–Stokes solve on noisy quantum hardware via a hybrid quantum–classical scheme”. *Computers & Fluids* **288**, 106507. DOI: [10.1016/j.compfluid.2024.106507](https://doi.org/10.1016/j.compfluid.2024.106507).
- [P30] Wilfong, B., A. Radhakrishnan, H. A. Le Berre, T. Prathi, S. Abbott, and S. H. Bryngelson (2025). “Testing and benchmarking emerging supercomputers via the MFC flow solver”. *SC Workshops '25: Proceedings of the SC '25 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis*, 669–677. DOI: [10.1145/3731599.3767424](https://doi.org/10.1145/3731599.3767424).
- [P29] Wilfong, Benjamin, Anand Radhakrishnan, Henry Le Berre, Daniel J. Vickers, Tanush Prathi, Nikolaos Tselepidis, Benedikt Dorschner, Reuben Budiardja, Brian Cornille, Stephen Abbott, F. *Schäfer, and S. H. *Bryngelson (2025). “Simulating many-engine spacecraft: Exceeding 1 quadrillion degrees of freedom via information geometric regularization”. *Proceedings of SC '25: The International Conference for High Performance Computing, Networking, Storage and Analysis*. *Equal contribution, 14–24. DOI: [10.1145/3712285.3771783](https://doi.org/10.1145/3712285.3771783).
- [P28] Zhu, Z., S. Remillard, B. A. Abeid, D. Frolkin, S. H. Bryngelson, J. Yang, M. Rodriguez Jr., and J. B. Estrada (2025). “Parsimonious inertial cavitation rheometry via bubble collapse time”. *Soft Matter* **21** 34, 6717–6734. DOI: [10.1039/D5SM00397K](https://doi.org/10.1039/D5SM00397K).
- [P27] Bati, A. and S. H. Bryngelson (2024). “RoseNNA: A performant, portable library for neural network inference with application to computational fluid dynamics”. *Computer Physics Communications* **296**, 109052. DOI: [10.1016/j.cpc.2023.109052](https://doi.org/10.1016/j.cpc.2023.109052).
- [P26] *Bryngelson, S. H., F. *Schäfer, J. Liu, and A. Mani (2024). “Fast Macroscopic Forcing Method”. *Journal of Computational Physics* **499**. *Equal contribution, 112721. DOI: [10.1016/j.jcp.2023.112721](https://doi.org/10.1016/j.jcp.2023.112721).
- [P25] Kocherla, S., Z. Song, F. E. Chrit, B. Gard, E. F. Dumitrescu, A. Alexeev, and S. H. Bryngelson (2024). “Fully quantum algorithm for mesoscale fluid simulations with application to partial differential equations”. *AVS Quantum Science* **6**, 033806. DOI: [10.1116/5.0217675](https://doi.org/10.1116/5.0217675).

- [P24] Liu, J., F. Schäfer, S. H. Bryngelson, T. A. Zaki, and A. Mani (2024). “Adjoint-based computation of nonlocal eddy viscosity in turbulent channel flow”. *Physical Review Fluids* **9**, 094606. DOI: [10.1103/PhysRevFluids.9.094606](https://doi.org/10.1103/PhysRevFluids.9.094606).
- [P23] Radhakrishnan, A., H. Le Berre, B. Wilfong, J.-S. Spratt, M. Rodriguez Jr., T. Colonius, and S. H. Bryngelson (2024). “Method for portable, scalable, and performant GPU-accelerated simulation of multiphase compressible flow”. *Computer Physics Communications* **302**, 109238. DOI: [10.1016/j.cpc.2024.109238](https://doi.org/10.1016/j.cpc.2024.109238).
- [P22] Sinha, A. and S. H. Bryngelson (2024). “Neural networks can be FLOP-efficient integrators of 1D oscillatory integrands”. *Transactions on Machine Learning Research*. ISSN: 2835-8856.
- [P21] Wilfong, B., A. Radhakrishnan, H. A. Le Berre, S. Abbott, R. D. Budiardja, and S. H. Bryngelson (2024). “OpenACC offloading of the MFC compressible multiphase flow solver on AMD and NVIDIA GPUs”. *SC24-W: Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis*, 1923–1933. DOI: [10.1109/SCW63240.2024.00242](https://doi.org/10.1109/SCW63240.2024.00242).
- [P20] Bryngelson, S. H., R. O. Fox, and T. Colonius (2023). “Conditional moment methods for polydisperse cavitating flows”. *Journal of Computational Physics* **477**, 111917. DOI: [10.1016/j.jcp.2023.111917](https://doi.org/10.1016/j.jcp.2023.111917).
- [P19] Elwasif, W., S. Bastrakov, S. H. Bryngelson, M. Bussmann, S. Chandrasekaran, F. Ciorba, M. A. Clark, A. Debus, W. Godoy, N. Hagerty, J. Hammond, D. Hardy, J. A. Harris, O. Hernandez, B. Joo, S. Keller, P. Kent, H. Le Berre, D. Lebrun-Grandie, E. McCarthy, V. G. Melesse Vergara, B. Messer, R. Miller, S. Oral, J.-G. Piccinali, A. Radhakrishnan, O. Simsek, F. Spiga, K. Steiniger, J. Stephan, J. E. Stone, C. Trott, R. Widera, and J. Young (2023). “Early application experiences on a modern GPU-accelerated Arm-based HPC platform”. *HPC Asia '23. International Workshop on Arm-based HPC: Practice and Experience (IWAHPCE)*. Singapore, 35–49. DOI: [10.1145/3581576.3581621](https://doi.org/10.1145/3581576.3581621).
- [P18] Firouznia, M., S. H. Bryngelson, and D. Saintillan (2023). “A spectral boundary integral method for simulating electrohydrodynamic flows in viscous drops”. *Journal of Computational Physics* **489**, 112248. DOI: [10.1016/j.jcp.2023.112248](https://doi.org/10.1016/j.jcp.2023.112248).
- [P17] Panchal, A., S. H. Bryngelson, and S. Menon (2023). “A seven-equation diffused interface method for resolved multiphase flows”. *Journal of Computational Physics* **475**, 111870. DOI: [10.1016/j.jcp.2022.111870](https://doi.org/10.1016/j.jcp.2022.111870).
- [P16] Zeng, Q., Y. Kothari, S. H. Bryngelson, and F. Schäfer (2023). “Competitive physics informed networks”. *International Conference on Learning Representations (ICLR)*. arXiv:2204.11144. Kigali, Rwanda.
- [P15] Charalampopoulos, A., S. H. Bryngelson, T. Colonius, and T. P. Sapsis (2022). “Hybrid quadrature moment method for accurate and stable representation of non-Gaussian processes and their dynamics”. *Philosophical Transactions of the Royal Society A* **380** 2229. DOI: [10.1098/rsta.2021.0209](https://doi.org/10.1098/rsta.2021.0209).
- [P14] Bryngelson, S. H., K. Schmidmayer, V. Coralic, K. Maeda, J. Meng, and T. Colonius (2021). “MFC: An open-source high-order multi-component, multi-phase, and multi-scale compressible flow solver”. *Computer Physics Communications* **266**, 107396. DOI: [10.1016/j.cpc.2020.107396](https://doi.org/10.1016/j.cpc.2020.107396).
- [P13] Spratt, J.-S., M. Rodriguez, K. Schmidmayer, S. H. Bryngelson, J. Yang, C. Franck, and T. Colonius (2021). “Characterizing viscoelastic materials via ensemble-based data assimilation of bubble collapse observations”. *Journal of the Mechanics and Physics of Solids* **152**, 104455. DOI: [10.1016/j.jmps.2021.104455](https://doi.org/10.1016/j.jmps.2021.104455).
- [P12] Bryngelson, S. H., A. Charalampopoulos, T. P. Sapsis, and T. Colonius (2020). “A Gaussian moment method and its augmentation via LSTM recurrent neural networks for the statistics of cavitating bubble populations”. *International Journal of Multiphase Flow* **127**, 103262. DOI: [10.1016/j.ijmultiphaseflow.2020.103262](https://doi.org/10.1016/j.ijmultiphaseflow.2020.103262).
- [P11] Bryngelson, S. H. and T. Colonius (2020). “Simulation of humpback whale bubble-net feeding models”. *Journal of the Acoustical Society of America* **147** 2, 1126–1135. DOI: [10.1121/10.0000746](https://doi.org/10.1121/10.0000746).
- [P10] Bryngelson, S. H., T. Colonius, and R. O. Fox (2020). “QBMMlib: A library of quadrature-based moment methods”. *SoftwareX* **12**, 100615. DOI: [10.1016/j.softx.2020.100615](https://doi.org/10.1016/j.softx.2020.100615).
- [P9] Schmidmayer, K., S. H. Bryngelson, and T. Colonius (2020). “An assessment of multicomponent flow models and interface capturing schemes for spherical bubble dynamics”. *Journal of Computational Physics* **402**, 109080. DOI: [10.1016/j.jcp.2019.109080](https://doi.org/10.1016/j.jcp.2019.109080).

- [P8] Trummler, T., S. H. Bryngelson, K. Schmidmayer, S. J. Schmidt, T. Colonius, and N. A. Adams (2020). “Near-surface dynamics of a gas bubble collapsing above a crevice”. *Journal of Fluid Mechanics* **899**, A16. DOI: [10.1017/jfm.2020.432](https://doi.org/10.1017/jfm.2020.432).
- [P7] Bryngelson, S. H. and J. B. Freund (2019). “Non-modal Floquet stability of a capsule in large amplitude oscillatory extension”. *European Journal of Mechanics B/Fluids* **77**, 171–176. DOI: [10.1016/j.euromechflu.2019.04.012](https://doi.org/10.1016/j.euromechflu.2019.04.012).
- [P6] Bryngelson, S. H., F. Guéniat, and J. B. Freund (2019). “Irregular dynamics of cellular blood flow in a model microvessel”. *Physical Review E* **100**, 012203. DOI: [10.1103/PhysRevE.100.012203](https://doi.org/10.1103/PhysRevE.100.012203).
- [P5] Bryngelson, S. H., K. Schmidmayer, and T. Colonius (2019). “A quantitative comparison of phase-averaged models for bubbly, cavitating flows”. *International Journal of Multiphase Flow* **115**, 137–143. DOI: [10.1016/j.ijmultiphaseflow.2019.03.028](https://doi.org/10.1016/j.ijmultiphaseflow.2019.03.028).
- [P4] Bryngelson, S. H. and J. B. Freund (2018). “Floquet stability analysis of capsules in viscous shear flow”. *Journal of Fluid Mechanics* **852**, 663–677. DOI: [10.1017/jfm.2018.574](https://doi.org/10.1017/jfm.2018.574).
- [P3] Bryngelson, S. H. and J. B. Freund (2018). “Global stability of flowing red blood cell trains”. *Physical Review Fluids* **3**, 073101. DOI: [10.1103/PhysRevFluids.3.073101](https://doi.org/10.1103/PhysRevFluids.3.073101).
- [P2] Bryngelson, S. H. and J. B. Freund (2016). “Buckling and its effect on the confined flow of a model capsule suspension”. *Rheologica Acta* **55** 6, 451–464. DOI: [10.1007/s00397-015-0900-9](https://doi.org/10.1007/s00397-015-0900-9).
- [P1] Bryngelson, S. H. and J. B. Freund (2016). “Capsule-train stability”. *Physical Review Fluids* **1** 3, 033201. DOI: [10.1103/PhysRevFluids.1.033201](https://doi.org/10.1103/PhysRevFluids.1.033201).